

AES IntelliNet 7744F / 7788F

NETCON 6/7

Signal Survey and Antenna Guide

Routing-table analysis, RX/TX/Monitor All diagnostics,
site surveying, antenna choice, coax and RF testing

FIELD OBJECTIVE

Determine whether poor NETCON is caused by local RF, antenna/coax, interference, subscriber configuration or weak upstream mesh routes.

Select the lowest-gain antenna and shortest practical coax that produces stable fire-compliant routing and verified central-station delivery.

1. Scope, safety and acceptance target

This guide is for diagnosing persistent NETCON 6 or 7 on AES IntelliNet 7744F and 7788F legacy fire subscribers. It uses the subscriber's Local Status, Routing Table, RX Monitor, TX Monitor and Monitor All functions together with controlled antenna-location and RF-path tests.

Fire-system target

The legacy subscriber manuals describe NETCON 0-6 as generally usable and 7 as a possible problem, but require fire installations to achieve NETCON 0-5. Current AES antenna-placement guidance likewise targets NETCON 5 for fire. A constant 6 or 7 on a 7744F/7788F fire subscriber requires investigation.

Safety requirements

- Notify the central station and place the account on test before reset, keying or event tests.
- Never key the transmitter without a correct 50-ohm antenna, dummy load or approved RF test instrument connected.
- Keep antennas and installation equipment away from electrical conductors. AES recommends at least 10 feet from external power lines.
- Use fall protection, lightning safety and qualified personnel for rooftop work.
- Remote outdoor antennas require approved surge protection, weatherproofing, grounding and code-compliant installation.
- Do not change subscriber frequency, power, cipher or network engineering parameters without authorization.

What constitutes a good final result

Check	Preferred field result
Local Status	Correct ID, STAT 000 or fully explained condition, valid RT1, LEVEL not 255
NETCON	0-5 for 7744F/7788F fire service
Routing table	Stable routes; preferably more than one useful path; Q02/Q03 routes available
WA indicator	Off after successful enrollment and acknowledgement
Packet behavior	Transmissions followed by acknowledgements without persistent retrying
Central station	Every required test event received and interpreted correctly

Guide sequence

Do not begin by installing the largest antenna. First determine whether the weakness is local RF, the feed line, interference, configuration or the upstream mesh. Then change one variable at a time and record the result after the routing table has had time to rebuild.

2. NETCON is not a signal-strength meter

NETCON is the subscriber's internal Network Connectivity rating. It reflects confidence in available paths through the mesh, not a direct measurement in dBm. Antenna work can improve NETCON when local RF links are poor, but an antenna cannot repair unhealthy routes farther upstream.

Observation	Likely implication
NETCON 6/7 with Q81/Q82/Q83 routes	Local RF, placement, antenna, coax or interference is a strong suspect
NETCON 6/7 with several Q03 routes	Local RF may be good; investigate neighbors' N/LEVEL and upstream mesh health
RT1:XXXX or LEVEL:255	Subscriber is not enrolled; check cipher, frequency, identity and usable network coverage
Route table nearly empty	Insufficient audible peers, severe attenuation, wrong channel/cipher or receiver-path problem
RT1 changes repeatedly	Unstable RF, intermittent interference, poor acknowledgements or unstable neighbors
Good status outside but poor inside	Building attenuation or indoor placement problem
Poor outside and inside	Network coverage, radio/coax fault or need for an engineered external antenna solution

NETCON interpretation

NETCON	Legacy-manual interpretation	Field use for these fire subscribers
0-4	Increasingly healthy as the number decreases	Normally favorable; still verify routes and central delivery
5	Acceptable and required maximum for fire	Survey target
6	Generally usable in legacy description	Not fire-compliant target; improve or escalate
7	May indicate a problem in the unit or its route	Unacceptable; isolate cause

Why a good antenna can still leave NETCON at 6

- The best-heard neighbors may themselves have high NETCON or high link levels.
- All usable neighbors may depend on the same upstream path, providing little route diversity.
- A neighboring repeater may have a fault, weak onward RF path or be leaving the network.
- The local unit may hear packets well but acknowledgements do not reliably return.
- Mesh density near the site may be inadequate even though one strong peer is audible.

Core diagnostic rule

Use Q to judge the local RF link to each routing-table peer. Use the peer's N and L values to judge the quality of the route beyond that peer. Use packet acknowledgements to confirm end-to-end behavior.

3. Capture a baseline before changing hardware

Reset clears monitor modes and forces a fresh enrollment/routing sequence. Record the initial condition first whenever possible.

Baseline procedure

Step	Action	Record
1	Confirm account is on test	Operator/time/contact
2	Inspect antenna, coax, connectors, power and board LEDs	Visible defects, water, corrosion, loose/crushed cable
3	Press physical Reset	SELFTEST result and WA behavior
4	Allow as long as 5 minutes for a stable NETCON/table	NETCON at 1, 3 and 5 minutes
5	Press d	ID, RT1, LEVEL, STAT, NETCON
6	Press Ctrl+T	Every route: preference, ID, L, N and Q
7	Press a, then c; add b if needed	Unique IDs heard, packet/acknowledgement behavior, retries
8	Turn monitors off or reset	Normal operating state restored

Status command

```
d
SUB [revision] 7788F
ID#:4923 (C)YYYY AES
RT1:1234 LEVEL:002
STAT:000 NETCON:6
```

Field	What to look for
RT1	Stable preferred next hop. XXXX means not on network.
LEVEL	Expected hops to IP-Link/receiver. 255 means not on network.
STAT	Correct power, battery, loopback, EEPROM, charger and ground-fault issues before judging RF.
NETCON	Use only after the table has had time to rebuild.

LED clues

LED	Diagnostic meaning
WA	On: waiting for acknowledgement. Blinking: not on network. Off: normal.
TX	Radio transmitter active.
RX	Receiver sees RF energy. Steady for about 20 seconds or more can indicate interference.
ALM	Normal steady blink pattern or a fault pattern for battery/zone/self-test conditions.

4. Read the routing table as an RF survey instrument

4.5678, L:02, N:5, Q:83

3.7888, L:01, N:5, Q:03

2.BB12, L:00, N:0, Q:03

1.AA11, L:00, N:0, Q:03

Field	Meaning	Survey use
Preference	1 is the current first-choice route	Watch for constant reshuffling or one-route dependence
ID	Subscriber or IP-Link next hop	Count unique viable peers and identify recurring weak paths
L	Peer's Link Layer / LEVEL	Same or lower than the local unit is normally desirable
N	Peer's NETCON	Shows upstream route confidence beyond the local RF hop
Q	Local received carrier/data quality	Primary local RF comparison between antenna locations

Decode Q correctly

Q is a two-character quality code, not an RSSI or dBm reading. The first character indicates whether Carrier Detect was active during reception. The second character indicates how many of the three repeated data copies were decoded.

Q	Carrier/data interpretation	Field rating
03	Carrier Detect on; all three copies decoded	Best
02	Carrier Detect on; two copies decoded	Good
01	Carrier Detect on; one copy decoded	Degraded
83	Carrier Detect not solid; all three copies decoded	Usable data but weak carrier indication
82	Carrier Detect not solid; two copies decoded	Marginal
81	Carrier Detect not solid; one copy decoded	Poor

A useful route

AES site-survey guidance describes a good route as a peer at the same or lower Link Layer, with NETCON 5 or 6 and Q02 or Q03. For the local 7744F/7788F fire subscriber, the completed installation should still achieve local NETCON 5 or better.

Routing patterns and conclusions

Pattern	Conclusion
One Q03 route, no alternatives	Strong local link but weak redundancy; antenna gain alone may not create mesh density
Several Q8x routes	Move antenna, improve height/clearance, inspect coax or test outdoors
Several Q03 routes with N6/7	Local RF is likely adequate; escalate upstream network optimization
Q alternates 03 to 81	Intermittent obstruction, interference, loose coax/connector or moving RF environment
Best Q improves outside	Building penetration is the limiting factor; remote antenna is justified

5. Use RX, TX and Monitor All to verify the path

Enable monitors in the correct order

- a -> RX MONITOR ON
- c -> MONITOR ALL ON
- b -> TX MONITOR ON, when local transmit visibility is required

Monitor All requires RX Monitor to be ON. These modes display decoded AES packets, not every RF signal that activates the RX LED. Monitor modes prevent incoming text messages; turn them off when finished or press Reset.

Packet format

<1234 #1A-> 0000 1234 -> FF01 <D_CHKIN>>

Field	Meaning
ORIG 1234	Subscriber that originally created the packet
PN 1A	Packet sequence number from 00 through FF
DEST 0000	Final destination, commonly the central receiver
FROM 1234	Unit currently transmitting the packet
TO FF01	Immediate next hop
D_CHKIN	Packet type; this example is check-in data

Packets useful during troubleshooting

Packet	Meaning	How to use it
D_CHKIN	Data check-in	Confirm origin, selected next hop and network activity
P_ACK	Packet acknowledgement	Verify that an outbound packet received a return acknowledgement
P_RR	Receiver Ready broadcast	Observe peers maintaining routing awareness

Monitor conclusions

Observation	Likely interpretation
TX packet followed quickly by matching P_ACK	The selected route completed a useful two-way exchange
Repeated TX packet number with no P_ACK	Acknowledgement path failure, weak route, congestion or remote peer issue
WA remains on after TX	Subscriber is waiting for acknowledgement
Monitor All hears many IDs but table remains poor	Traffic is audible, but peers may not qualify as good routes
RX LED active but no decoded lines	Interference, other protocol, wrong frequency traffic or undecodable AES signal
No RX LED and no Monitor All traffic	No audible RF energy; inspect receiver path, location, antenna and network coverage

6. Decision process for constant NETCON 6/7

Decision	If YES	If NO
STAT is 000 or the displayed condition is fully explained?	Continue to enrollment check	Correct power, battery, loopback, charger or ground fault first
RT1 is valid and LEVEL is not 255?	Continue to routing-table analysis	Verify ID, cipher, frequency, antenna and network availability
At least two routes have Q02/Q03?	Judge peers' N and L values	Treat as local RF/placement/feed-line problem
Peers' N and L indicate healthy onward paths?	Analyze acknowledgements and stability	Escalate upstream mesh density/neighbor health
Packets receive consistent P_ACK responses?	Repeat survey at alternate locations	Investigate chosen route, interference and two-way RF balance
Outside free-air test reaches NETCON 5?	Building attenuation; design remote antenna	Test higher antenna/gain and evaluate network coverage
Wattmeter/coax tests pass?	Antenna location/network design remains	Repair radio, coax, connector, surge protector or antenna

Root-cause signatures

Root cause	Typical signature
Indoor attenuation	Outside Q/NETCON clearly better than inside; metal/low-grade location
Bad coax/connector	Results worsen when remote cable is added; reflected power high; movement changes reading
Wrong antenna/frequency	Poor Q everywhere; no stable improvement from placement
Interference	RX LED stays active, decoded traffic sparse, Q unstable across time
Weak upstream mesh	Local routes Q03 but peer N/LEVEL poor and local NETCON remains high
Insufficient local peers	Routing table has one or very few entries even in clear outdoor location
Wrong cipher/channel	No enrollment, RT1 XXXX, LEVEL 255, useful network packets absent
Low transmit performance	Others audible, but local packets lack P_ACK; wattmeter or supply test fails

Do not solve every NETCON problem with gain

If multiple peers are already Q03 but their N and L values are poor, more antenna gain can create overreach without repairing the upstream network. Document the table and involve the network administrator.

7. Isolate antenna/feed line, transceiver or network signal

Use controlled substitution. Change only one part at a time and use the same location, wait interval and routing-table capture. A continuity test alone cannot prove that an antenna or coax works correctly at RF.

Three controlled reference tests

Reference test	What it isolates
Radio into known-good 50-ohm load through wattmeter	Transmitter output and power supply without the installed antenna/feed line
Original subscriber with known-good short cable and matched antenna	Installed coax, surge protector, adapters and antenna
Known-good compatible subscriber/transceiver on same authorized frequency at same antenna location	Original transceiver receive/transmit performance versus external network conditions
Control conditions A comparison transceiver must be authorized, frequency-compatible and connected under central-station test procedures. Do not introduce a duplicate active subscriber ID. Record which board, transceiver, ID and antenna are used for every trial.	

Antenna or feed-line fault signature

- Forward power is normal into a known-good dummy load.
- Reflected power is excessive only when the installed feed line/antenna is connected.
- A known-good short cable and matched antenna produce better Q and NETCON at the same location.
- Results change when a connector or cable is moved.
- Water, corrosion, loose connectors, braid shorts, crushed coax or an incorrect-frequency antenna is found.
- The case-top antenna works, but adding the remote coax/surge/antenna makes the routing table worse.

Transceiver or power fault signature

- Forward power is low into a known-good 50-ohm load.
- Supply voltage drops abnormally while e keys the transmitter.
- Loopback/status fault 080 persists after reset.
- With the same known-good antenna and location, a compatible known-good radio hears substantially more IDs or better Q.
- Local TX packets repeatedly receive no P_ACK while a comparison unit at the same location succeeds.
- Transmit power can be normal while receiver sensitivity is poor; compare Monitor All and routing tables between radios.

7. Isolation decision table

Test result	Most likely category	Next action
Low forward power into known-good load	Transceiver, RF connector or DC supply	Verify DC under key; inspect radio connection; service/replace authorized hardware
Normal forward; reflected >10% on installed system	Antenna/feed line mismatch	Move wattmeter down the path and isolate cable, protector, connector or antenna
Installed system poor; known-good short antenna good	Permanent antenna/coax/mount	Repair feed line or select/install correct antenna
Original radio poor; known-good compatible radio good on same antenna	Original transceiver/receiver path	Confirm supply and connectors, then service/replace transceiver
Both radios poor on known-good antenna outside	External coverage, interference or upstream mesh	Review Monitor All, route N/L and involve network administrator
Both hear many Q03 peers but NETCON remains 6/7	Upstream route quality/topology	Document peer N/L values; do not add gain blindly
RX LED steady; both radios decode little traffic	Interference/noise at site	Use service monitor, change location and investigate interference source
Good RX/Monitor All; local TX lacks ACK; forward power good	Two-way imbalance or peer/network issue	Compare routes and P_ACK behavior; test alternate location/route

Fast A/B sequence

Order	Configuration	Interpretation
A	Original radio + installed feed/antenna	Baseline
B	Original radio + known-good short feed/antenna	Improvement identifies installed feed/antenna problem
C	Original radio + known-good short feed/antenna outside	Improvement over B identifies building attenuation/location
D	Known-good compatible radio + same known-good outside antenna	Improvement over C identifies original transceiver/power path
E	Both radios poor in D conditions	External network signal, interference or upstream topology

Receiver comparison using Monitor All

At the same antenna location, reset each radio, enable a then c, use the same capture interval and count unique decoded IDs. Compare the full routing table and Q values. A materially weaker result from one radio under identical conditions points to its receiver path. Similar weak results from both radios point away from the transceiver and toward location, interference or lack of network coverage.

Do not condemn a receiver from RX LED behavior alone
The RX LED indicates RF energy, including noise. Receiver performance must be compared by decoded packets, route-table Q, and a known-good radio/antenna control.

8. Controlled site-survey procedure

Phase A - outside free-air reference

Before committing to an indoor location, establish whether viable mesh coverage exists outside the facility. AES notes that a poor result outside will not improve inside the building.

- Use the correct subscriber ID and network cipher.
- Begin with the case-top/rubber-duck antenna or a 7214-EMK test mount.
- Keep the temporary antenna clear of vehicles, large metal surfaces and the technician's body.
- Reset after every move and allow up to 5 minutes for the final routing table and NETCON.
- Record d status, the complete Ctrl+T table and a short a/c Monitor All sample.
- If NETCON 5 cannot be reached outside, test an approved higher-gain antenna before quoting a normal indoor installation.

Phase B - indoor candidate locations

Use the 7214-EMK with the rubber-duck antenna to create the needed mounting/radial condition and move the antenna through candidate locations using an appropriate 6-, 10- or 25-foot test cable.

- Test higher locations first.
- Prefer an outside wall and a clear view toward the expected network coverage area.
- Avoid basements, below-grade rooms, elevator equipment, large metal cabinets and metal roof shielding.
- Keep the antenna separated from electrical wiring, motors, transformers and switching power equipment.
- Reset after each move. Do not compare a newly moved antenna against a previous fully developed table.

Phase C - make each measurement comparable

Item	Keep constant
Subscriber	Same radio, ID, cipher, power source and firmware
Antenna trial	Change only location first; then gain after location has been optimized
Wait time	Use the same 5-minute stabilization interval
Data capture	Status, full table, monitor interval and WA behavior
Environment	Note doors, roof equipment, vehicles, occupancy, weather and known interference

Choose stability, not one lucky reading

A location that briefly shows NETCON 5 but repeatedly returns to 6/7 or loses its first route is not a strong installation.

Repeat measurements and favor stable route diversity.

9. Antenna placement

Indoor placement

- Use the case-top antenna where it produces compliant and stable routing.
- Use a proper mounting kit/ground-plane arrangement when remotely testing a rubber-duck antenna.
- Move higher and toward an outside wall before increasing antenna gain.
- Keep the radiating element away from large grounded metal, electrical wiring and noise sources.
- Do not coil excess coax beside the antenna or sharply bend/crush the feed line.

Outdoor placement

Current AES placement guidance recommends an unobstructed view of the intended coverage area, ideally 360 degrees. The antenna should be plumb and clear of walls, parapets and roof structures.

Placement item	AES field guidance
Roof height	Prefer the antenna/ground plane about 5-10 feet above the roofline where practical
Obstruction clearance	At least about 12 inches from parapets, walls and other obstructions
Metal roof	Keep the radiating element above the roofline; avoid transmitting through the building
Orientation	Mount the omnidirectional antenna vertical/plumb
Visibility	Seek 360-degree visibility to nearby subscribers
Power lines	Maintain at least 10 feet separation from external power lines
Weather	Physically protect and weatherproof outdoor connectors and cable

Height must not create a code or lightning hazard

Use approved mounts, grounding, bonding and surge protection. Roof work and antenna placement must comply with the National Electrical Code, local requirements and the authority having jurisdiction.

Common placement failures

Failure	Result
Antenna beside metal cabinet/duct	Pattern distortion, shielding and unstable Q
Element below metal roofline	Signal forced through/along building structure
Horizontal or tilted omni	Wrong polarization and uneven coverage
Antenna next to power equipment	Increased interference/noise risk
Outdoor connector unsealed	Water intrusion, corrosion and increasing reflected power
Long unnecessary coax	Feed-line loss consumes antenna gain

10. Choose the antenna by evidence

AES selection rule

Use the smallest antenna gain that produces a strong, stable connection. AES advises avoiding antennas above 6 dB on subscriber installations unless specifically engineered; bigger is not automatically better.

Antenna class	Typical use in survey	Selection guidance
2.5 dB case-top / rubber duck	First trial and preferred simple installation	Use when compliant NETCON and route diversity are stable
Phantom / 3 dB	Low-profile or modest remote improvement	Try after location optimization; confirm correct mounting/ground plane
5 dB omni	Additional outdoor/remote coverage	Use when measured routing improves and coax loss is controlled
6 dB omni	Higher-gain engineered subscriber option	Use only when smaller antennas cannot meet the documented target
7 or 9 dB omni	Longer-range/head-end or special design	Avoid as a default subscriber fix; require AES/network engineering approval

Relevant AES antenna families

Part / family	Description / note
7214 / 7214-EMK	2.5 dB rubber-duck/case or extension-mount survey arrangement
7264-PA-SUB	Remote phantom-antenna package
7210-3	3 dB omnidirectional UHF antenna
7210-5	5 dB omnidirectional UHF antenna
7210-6	6 dB omnidirectional UHF antenna
7210-7	7 dB option; current accessory guide identifies 460-470 MHz range only
7210-9	9 dB omnidirectional antenna; not a default legacy-subscriber choice

Antenna trial order

Order	Action	Advance only when
1	Case-top antenna in best practical subscriber location	NETCON/route target cannot be made stable
2	Same antenna moved using EMK/test cable	Best indoor position remains inadequate
3	Same antenna in outside free air	Outside reference remains inadequate or proves building loss
4	3 dB or phantom antenna	Measured table/Q improvement is insufficient
5	5 dB antenna with correct coax	Smaller gain cannot reach target
6	6 dB or engineered solution	Documented need and network/AES approval

Why too much gain can be harmful

Higher-gain omnidirectional antennas concentrate energy into a flatter vertical pattern. They can hear/transmit farther while reducing useful close-in vertical coverage, creating overreach and allowing distant traffic to dominate. Select gain from route-table evidence, not from the largest available dB number.

11. Coax, connectors, surge protection and weather

Cable requirements

AES requires 50-ohm RF cable. Use the shortest practical run and only compatible connectors and tools.

Cable	AES maximum / recommendation	Field use
RG-58	25 feet maximum including surge protector	Short subscriber-to-antenna runs
RG-8 / Belden 9913 equivalent	75 feet maximum	Longer low-loss runs with correct connectors
LMR-400	Current Antenna Matrix recommends 100 feet or less	Longer remote-antenna runs

Some older AES articles mention LMR-400 operation up to 125 feet. For new work, use the more conservative current Antenna Matrix recommendation of 100 feet or less unless AES engineering approves otherwise.

Inspection checklist

- Correct 50-ohm cable type and connector family.
- No sharp bends, crushing, staples through the jacket or pulled center conductor.
- No loose adapters, damaged threads, braid shorts or incomplete crimps.
- No water, green corrosion or blackened connector surfaces.
- Cable shield and center conductor pass continuity tests and are not shorted together.
- Feed line is not longer than necessary.

Outdoor installation

- Use AES 7230 or the currently approved surge protector for every remote outdoor antenna.
- Install grounding and bonding according to AES instructions, NEC and local code.
- Protect threaded connections with suitable dielectric grease.
- Seal exposed connectors with approved mastic/self-fusing weatherproofing tape.
- Provide drip loops and physical protection where required.

Loss-versus-gain reality

A higher-gain antenna at the end of excessive or defective coax can perform worse than a smaller antenna on a short feed. Compare the routing table before and after adding the complete cable, surge protector and adapters - not the antenna alone.

Useful isolation test

If a case-top antenna works but the remote antenna does not, temporarily substitute a known-good short cable and antenna. Then add the permanent feed-line sections one at a time to locate the loss.

12. RF power and reflected-power testing

Use the e key-transmitter function

```
e
KEYING TX..
TIMEOUT
```

The e command keys the transmitter for approximately five seconds for wattmeter, antenna and coax tests. Press Enter to abort early. TIMEOUT after the full test interval is normal. This is a carrier test and may not create a decoded TX Monitor packet.

Required equipment

- 50-ohm wattmeter rated for the subscriber frequency and power.
- Correct patch leads and connector adapters.
- Known-good 50-ohm dummy load and/or verified antenna system.
- DC meter suitable for checking supply behavior under transmit load.

AES reflected-power criterion

AES states that reflected power should be no more than 10 percent of forward power. Example: with 1.9 W forward, reflected power should not exceed 0.19 W.

Result	Interpretation	Next action
Forward power normal; reflected $\leq 10\%$	Radio and tested load path are likely acceptable	Continue route/location analysis
Forward power low into known-good load	Radio supply, radio, connector or test setup problem	Check DC under load and isolate radio output
Forward normal; reflected $>10\%$	Mismatch downstream of meter	Move meter to next connection point
Good at radio, bad after coax	Cable/connector/surge protector/antenna fault	Test each segment
Power changes when cable moves	Intermittent connector or damaged coax	Replace/repair and retest

Process of elimination

Test point	Purpose
Radio into known-good dummy load	Establish transmitter baseline
Radio plus short patch	Verify patch/adapters
Before surge protector	Evaluate indoor feed segment
After surge protector	Evaluate protector and grounding connection
At antenna feed point	Evaluate complete coax path
Complete installed antenna	Verify final match and output

Do not use e as proof of network delivery

A keyed transmitter and acceptable wattmeter reading prove RF hardware performance only. Final proof requires useful packets, acknowledgements, compliant routing and central-station event confirmation.

13. Survey interpretation matrix

Inside	Outside	Routing / monitor pattern	Conclusion / action
NC 6/7; Q8x	NC <=5; Q02/03	Acknowledgements improve outside	Building attenuation. Remote antenna outside using correct coax.
NC 6/7; Q8x	NC 6/7; Q8x	Sparse table/no ACK	Coverage, radio/feed fault or major interference. Try approved gain and RF tests.
NC 6/7; Q03	NC 6/7; Q03	Peers N6/7/high L	Upstream mesh problem; antenna gain is unlikely to repair it.
NC fluctuates	NC stable <=5	Q changes when cable/antenna moves	Indoor multipath or intermittent feed line. Relocate/repair.
RT1 XXXX; L255	Same outside	No useful AES traffic	Check ID/cipher/frequency and network availability before antenna changes.
Table full; Q03	Same	Repeated TX without P_ACK	Two-way imbalance, route/peer issue, congestion or weak local TX.
RX LED steady	Same	Monitor All mostly blank	Interference/noise. Use service monitor and relocate source/antenna.

When to escalate to the network administrator

- Multiple local Q03 routes exist but local NETCON remains 6/7.
- Every viable peer shows high N or high L values.
- The same weak upstream ID appears in many affected subscribers.
- A subscriber has only one viable route even in clear outdoor conditions.
- Packet acknowledgements are inconsistent across multiple known-good local installations.
- A frequency/interference survey suggests a network-wide channel problem.

Information to provide

- Subscriber ID, model, firmware, location and antenna/coax configuration.
- Status output at 1, 3 and 5 minutes after reset.
- Complete routing tables from inside, outside and final antenna location.
- Monitor lines showing packet numbers, FROM/TO IDs, retries and acknowledgements.
- Forward/reflected-power measurements and supply voltage under key.
- Photographs of antenna height, clearance, cable, surge protector and connectors.

Constant NETCON 6/7 with good Q is data

Do not discard this evidence. A full table of Q03 neighbors with poor N/LEVEL values demonstrates that the antenna is hearing well and the remaining problem is likely network topology or upstream health.

14. Field survey worksheet

Complete one row only after Reset and the same stabilization interval at each location.

Trial	Location / height	Antenna	Cable	NC	RT1	L	# routes	Best Q	Q02/03 count
1									
2									
3									
4									
5									
6									
7									
8									

Trial	WA after 5 min	RX activity	TX/P_ACK result	Unique IDs heard	Interference / notes
1					
2					
3					
4					
5					
6					
7					
8					

Final antenna/feed-line record

Field	Recorded value
Selected antenna / part	_____
Mount location / height	_____
Cable type / measured length	_____
Surge protector / grounding	_____
Forward / reflected power	_____
Final RT1 / LEVEL / STAT / NETCON	_____
Final route IDs and Q	_____
Central-station event confirmation	_____
Technician / date	_____

15. Final acceptance and closeout

Step	Acceptance check
1	Final antenna is secure, vertical, weatherproof and properly clear of obstructions/power lines.
2	Coax is correct 50-ohm type, within length limit, undamaged and properly terminated.
3	Outdoor surge protection, grounding and bonding are complete and code compliant.
4	Physical Reset completed; subscriber allowed up to 5 minutes to rebuild routing.
5	Status shows correct ID, acceptable STAT, valid RT1, LEVEL not 255 and NETCON 0-5.
6	Routing table contains stable useful peers, with Q02/Q03 routes where available.
7	Monitor testing shows useful packets and acknowledgements without persistent retries.
8	RX is not continuously held by interference.
9	Forward/reflected power meets AES criterion when RF testing was required.
10	Every required alarm, trouble and restoral was received correctly by the central station.
11	a, b and c monitors are OFF; account removed from test; final readings documented.

Quick diagnostic commands

PC key	Function	Use in this guide
d	Local Status	RT1, LEVEL, STAT and NETCON
Ctrl+T	Routing Table	Preference, ID, L, N and Q
a	RX Monitor	Packets addressed to this subscriber
c	Monitor All	All decoded AES traffic in range; requires a
b	TX Monitor	Packets transmitted by this subscriber
e	Key Transmitter	Five-second RF wattmeter/antenna test

Technician escalation summary

Local RF evidence: route Q values and inside/outside comparison
 Upstream evidence: neighbor N and L values
 Two-way evidence: TX packet followed by P_ACK
 Hardware evidence: forward/reflected power and coax isolation
 Final proof: central station receives correct events

Official AES sources

AES 7744F Installation and Operation Manual

<https://aes-corp.com/wp-content/uploads/2020/07/7744-Install-Manual.pdf>

AES 7788F Installation and Operation Manual

<https://aes-corp.com/wp-content/uploads/2020/07/40-7788-Rev-6.pdf>

AES Best Practices for Antenna Placement

<https://aes-corp.com/best-practices-for-antenna-placement/>

AES Site Survey

<https://aes-corp.com/aes-site-survey/>

AES Antenna Matrix Installation and Buying Guide, Rev 5

https://aes-corp.com/wp-content/uploads/2022/09/AES_Antenna-Matrix-Buying-Guide_Rev5.pdf

AES Coax Cable Selection and Termination

<https://aes-corp.com/coax-cable-selection-and-termination/>

AES How to Test RF Performance

<https://aes-corp.com/how-to-test-rf-performance/>

AES 5 Keys to Proper Installation and Network Optimization

<https://aes-corp.com/5-keys-to-proper-installation-network-optimization/>

Guidance differences

AES web guidance published at different times gives both minimum and preferred roof-clearance figures, and older articles mention LMR-400 runs up to 125 feet. This guide uses the stricter current placement recommendation and the Antenna Matrix recommendation of 100 feet or less. Follow the latest AES engineering direction for the specific installation.

This guide supplements the AES 7744F/7788F Complete Technician Guide. It does not replace the official installation manuals, central-station network policy, approved system drawings, listings or authority-having-jurisdiction requirements.